

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM AND LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

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Flying robots may soon clean your house

What if your house could be cleaned automatically, or you had a vacuum cleaner that could work on its own? Well, this may soon turn into a reality as Adrian Perez Zapata, industrial design student at Colombia, has designed a system, which cleans by itself using mini flying robots.

The 23-year-old has conceptualised an automated cleaning system, where flying robots will be cleaning your house. Known as Mab, the system uses a number of mini-robots,



Adrian Perez Zapata's automated cleaning system

which are fitted with cleaning solutions. Zapata's system has brought him amongst the 20 semi-finalists in the 2013 Electrolux Design Lab competition.

Zapata's concept needs users to do minimum bit of work. All one has to do is to put water and a particular chemical into the robot's spherical core. This is then spread amongst 908 mini-robots, which can fly using propellers. These robots are then set free in the house. They scan the house and clean dirty surfaces using droplets of the liquid. The mini robots are powered using solar energy.

The system can be scheduled to clean at particular times or particular areas of a house. It can also be synced with smartphones and controlled through them.

Wireless devices to communicate without battery

A team of researchers led by Indian-origin Shyam Gollakota from the University of Washington has devised a way for wireless devices to communicate without the need for batteries. Known as Ambient Backscatter, the technology uses wireless signals like television and cellphone transmissions surrounding us. According to reports, the wireless devices communicate by reflecting these signals and exchanging information through them.

Gollakota's team has designed batteryless wireless devices with attached antennae, which could detect, harness and then reflect television signals. These signals were then picked up by similar devices.

After further studies, this technology could turn out to be highly useful, as it would eliminate the need for a power source in wireless or network-related devices. This would also lead to less requirement of human attention towards such devices.